What 2 Play?

A Boardgame Recommender

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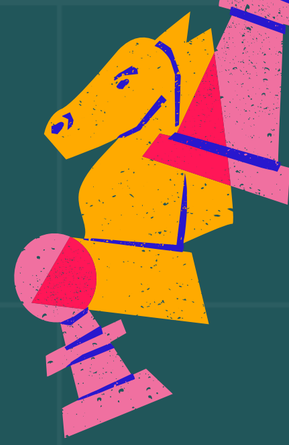
Project Overview

Goal: Build a personalized boardgame recommender

- Get Boardgame recommendations based on **popularity**
- Updated with personal **preferences**, allowing the app to be personalized for each user.
- Combining ranking **predictors with similarity** matrices.
- Allow the user to update the data with their **personal ratings.**
- With an **user-friendly** Streamlit App

Data Analysis



- **Two main datasets** gathering info from BGG
  - Most updated ranking
  - All kind of details of each game

- Almost **30.000 games** with more than 50 columns of details

- **Key decissions** for the data cleaning.
  - Lots of columns with information non relevant for this project pourpose

- **Keep it simple, keep it useful**

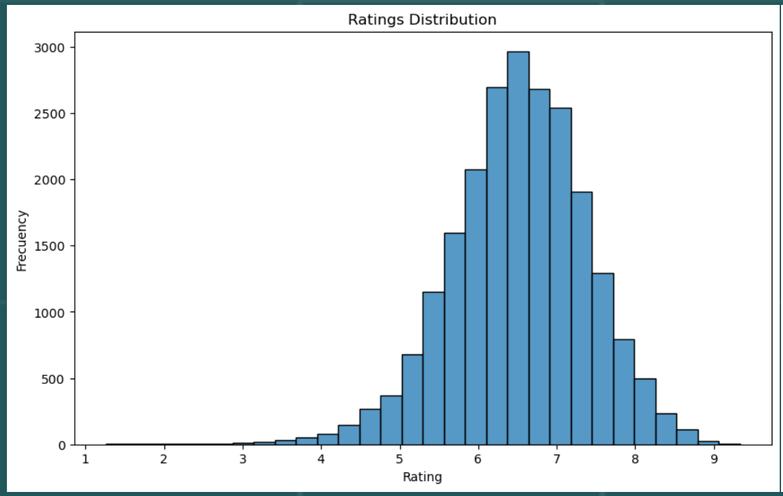
Two categorical columns with lists on them

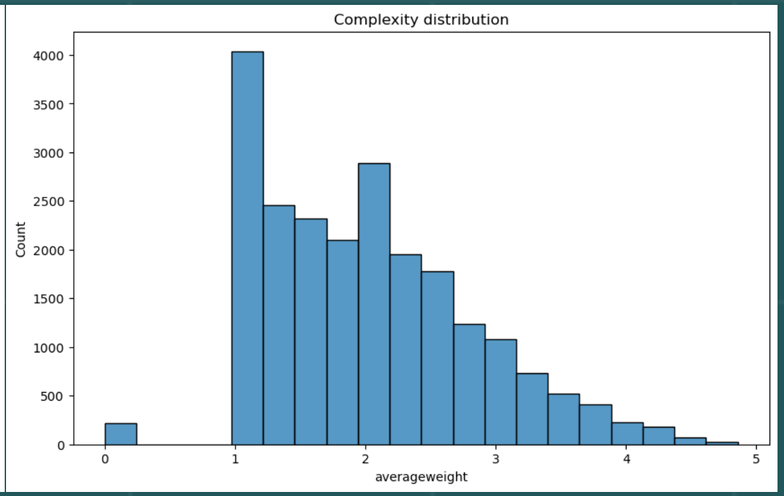
- Categories
- Mechanics

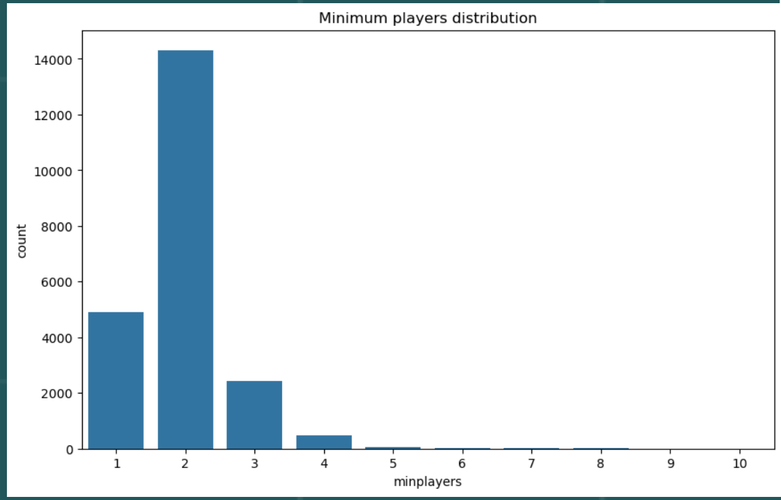
- Tokenization and TF-IDF applied for the feature engineeing

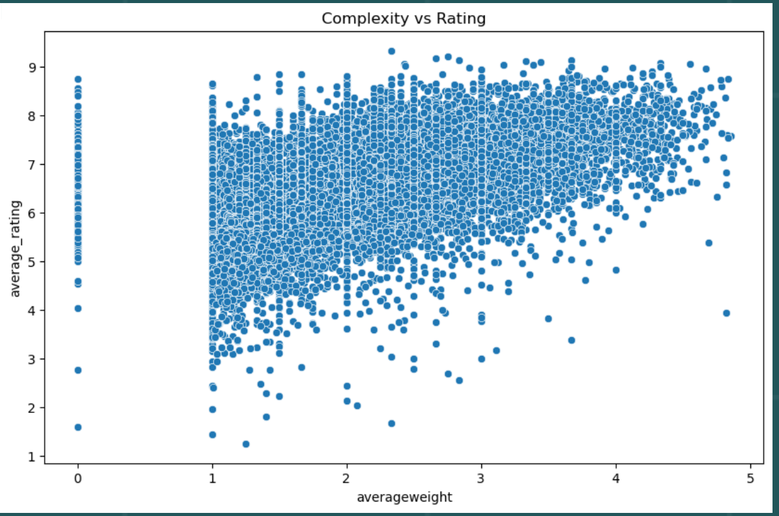
Data Analysis

Distribution of Data and Complexity plot



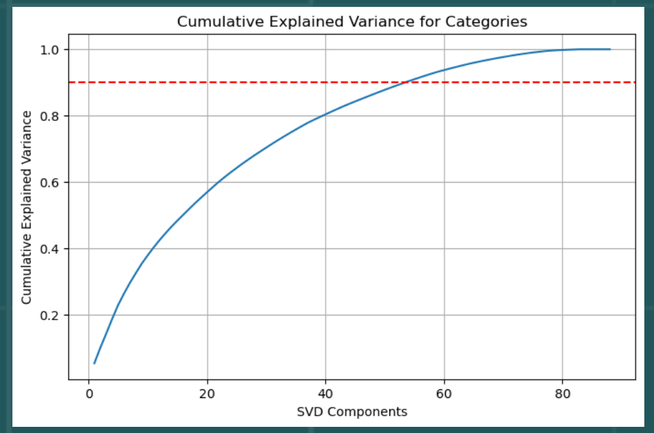


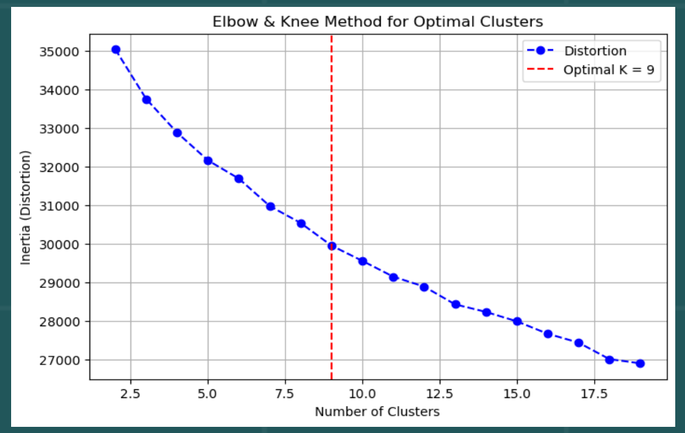


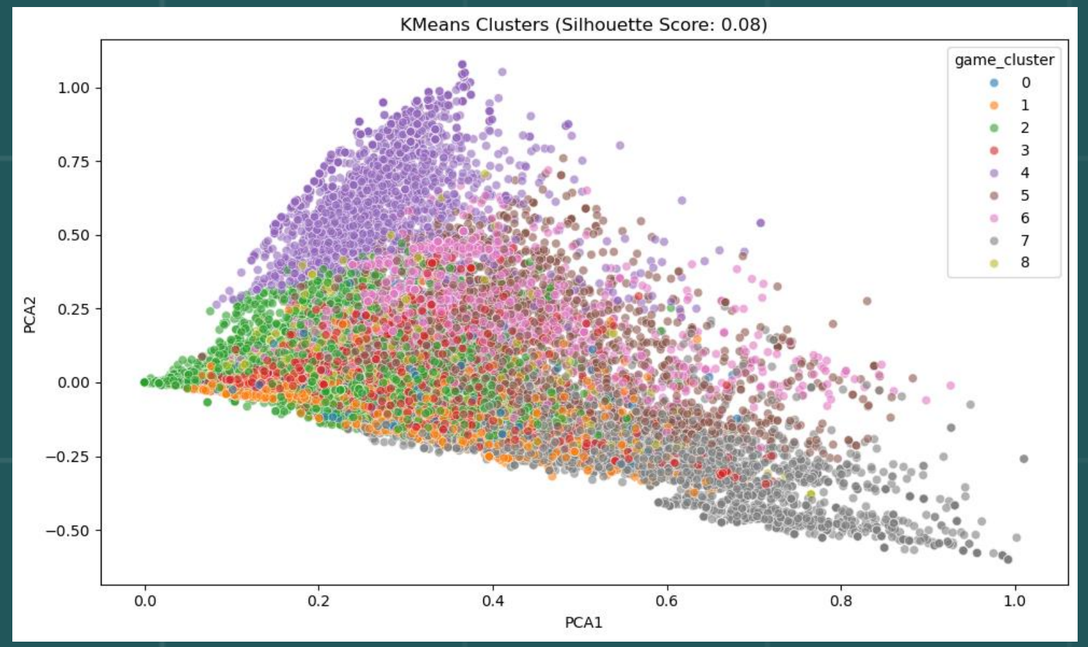


Feature Engineering

- **Logaritmic transformation**
- **New features**
  - Combining previous ones
- **TruncatedSVD**
  - Cumulative Experience Variance
  - Prefered over PCA due to the sparse data after TF-IDF
- **KMeans**
  - Elbow & Knee Method





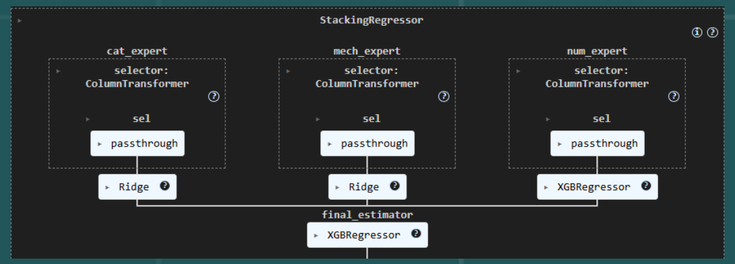


Model Training

- **6 models trained**
  - Linear Regression
  - Random Forest
  - Gradient Boosting
  - XGBoost
  - CatBoost
  - ExtraTrees
  - Ridge

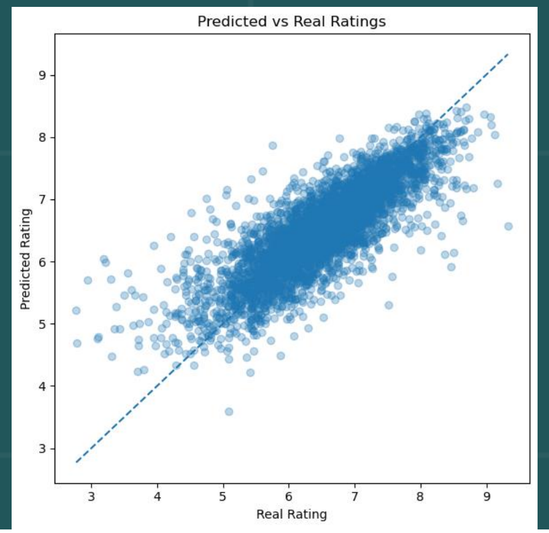
	Model	MSE	RMSE	R2	CV Mean
0	Linear Regression	0.315196	0.561424	0.581340	0.567359
1	Random Forest	0.264577	0.514370	0.648575	0.636956
2	Gradient Boosting	0.263833	0.513647	0.649564	0.637313
3	XGBoost	0.240993	0.490910	0.679901	0.664567
4	CatBoost	0.261688	0.511555	0.652412	0.642087
5	ExtraTrees	0.269205	0.518850	0.642428	0.631024

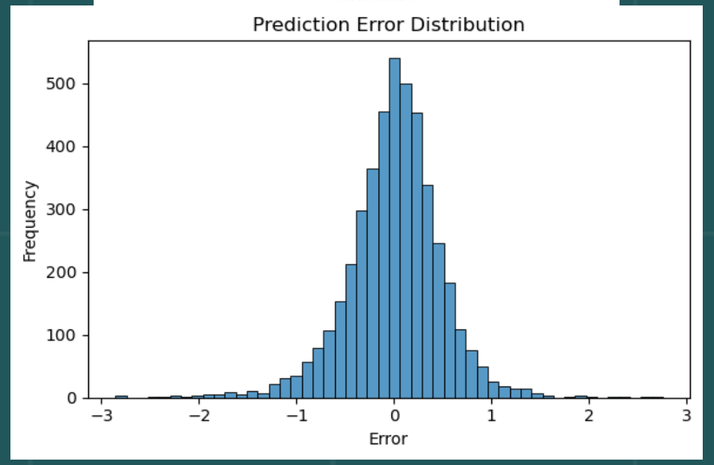
- **Hyperparameters**
- **Stacking**



Best model found

XGBoost with Hyperparameters



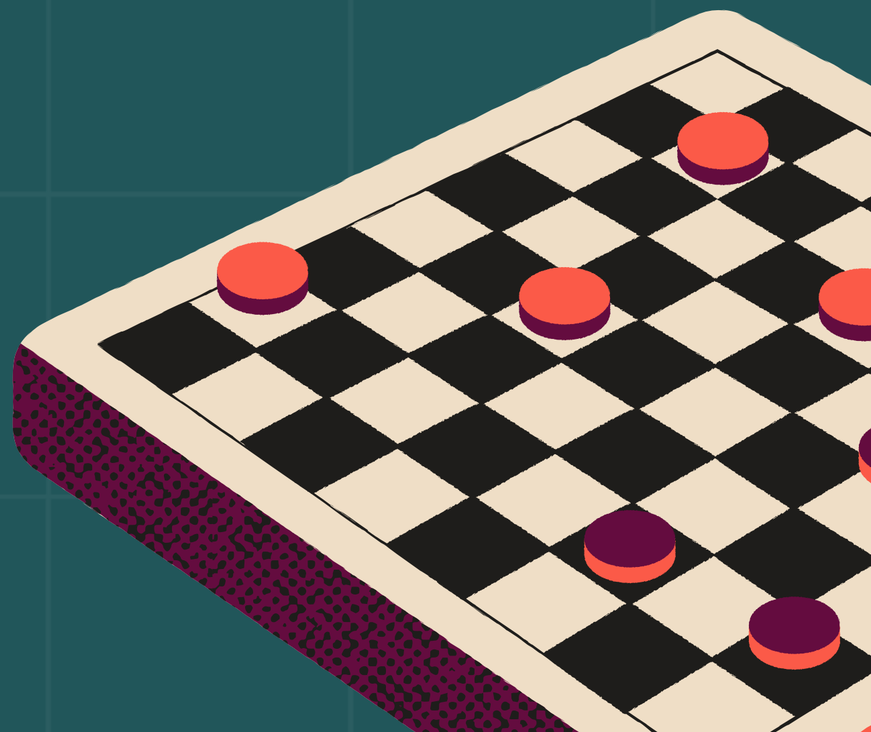


Recommender Function

- The **Key Code**
- **Similarity function**
  - Using cosine similarity
- **Popularity function**
  - Using the trained model to predict rating
- **Hybrid model**
  - 60% Popularity 40% Similarity

✓ Defined **4 features** that the user would have

✓ Filters for the model
  - Number of players
  - Duration of the game
  - Complexity
  - Age of the game

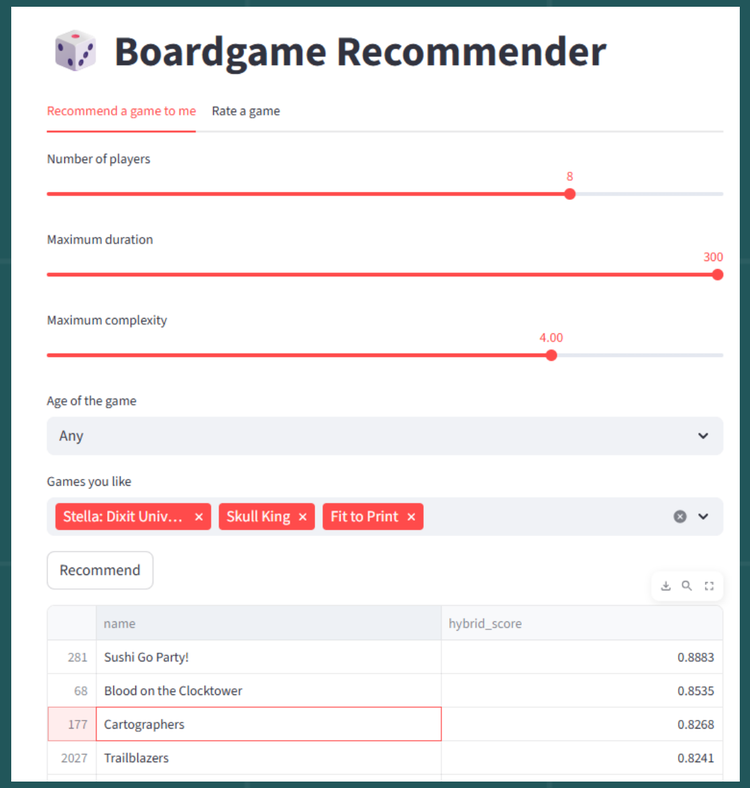


Streamlit App

Streamlit App

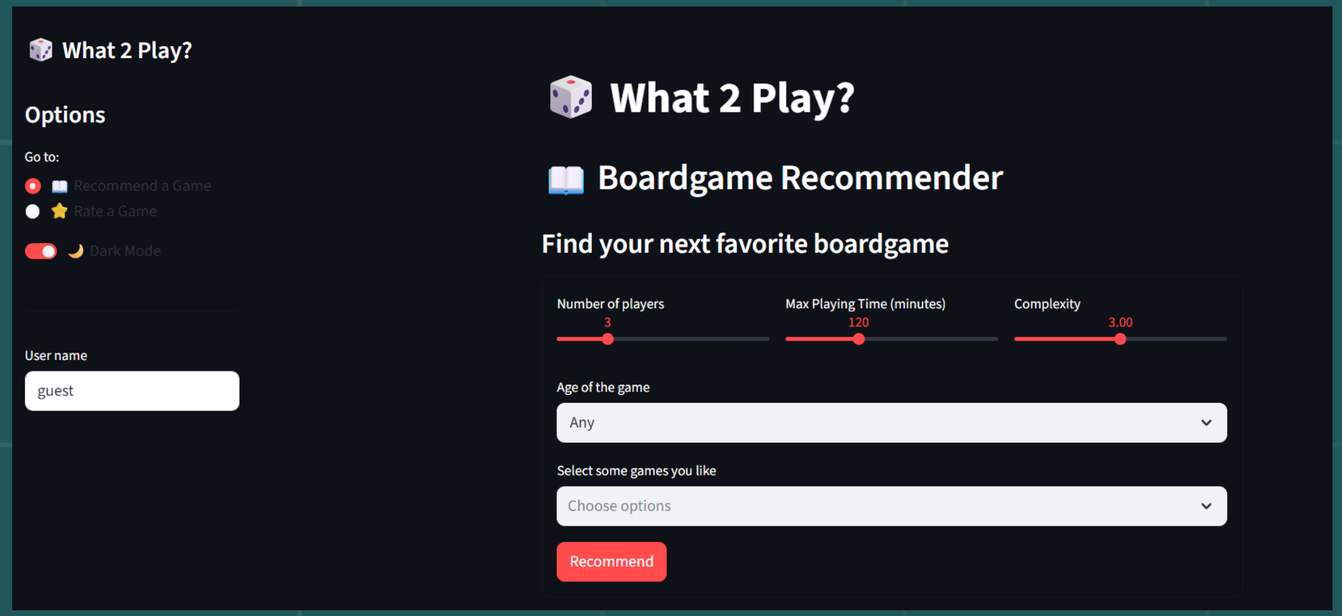
How it started

- Basic
- Ugly
- Ideal for modifications

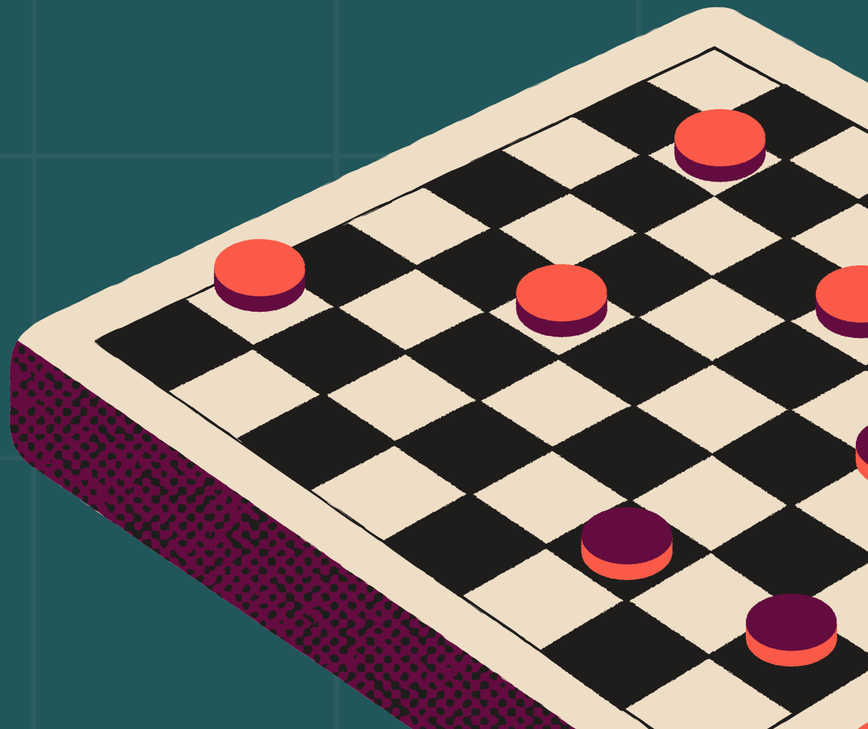


Improvements

- ✓ Prettier
- ✓ Faster
- ✓ Better organized



Demo Time!



Next Steps

- Update of the database with new game releases
  - Ratings could also change as per the nature of the data
  - ¿API to do it automatically?

- Add YouTube links on "How To Play" to add to each game card

- Connectivity beetween users
  - Share your favorite games and rating with your friends

Thank You